

Executive Summary

EXECUTIVE SUMMARY

Project Title:

Healthcare Capacity Analytics and Forecasting System

Overview:

The Unaccompanied Alien Children (UAC) Program is a federally mandated humanitarian care initiative managed by the U.S. Department of Health and Human Services (HHS). The program is responsible for the intake, medical support, sheltering, and reunification of unaccompanied children transferred from U.S. Customs and Border Protection (CBP).

This project presents an AI-powered Healthcare Capacity Analytics and Forecasting Dashboard designed to monitor operational care load, intake pressure, discharge effectiveness, and predictive capacity trends across the humanitarian healthcare pipeline.

Objectives:

- Monitor total healthcare system load across CBP and HHS
- Analyze inflow and outflow balance
- Detect periods of operational strain and backlog accumulation
- Forecast future healthcare system load using predictive analytics
- Support operational decision-making and humanitarian planning

Methodology:

The project uses historical operational data from 2023–2025 and applies:

- Data preprocessing and cleaning
- Time-series trend analysis
- KPI generation
- Rolling average monitoring
- Predictive forecasting using Facebook Prophet
- Interactive dashboard visualization using Streamlit and Plotly

Key Findings:

- HHS Care consistently carried the majority of the operational healthcare load.
- Elevated Net Intake Pressure indicated temporary operational stress periods.
- Rolling averages revealed sustained high-load intervals requiring monitoring.
- Forecasting results suggested gradual stabilization in projected care demand trends.

Dashboard Features:

- KPI summary cards
- CBP vs HHS comparative analysis
- Intake pressure monitoring
- Forecasting and predictive analytics
- Operational alerts and recommendations
- Interactive filters and date selection

Recommendations:

- Strengthen discharge coordination mechanisms
- Improve temporary shelter capacity planning

- Increase predictive monitoring for intake surges
- Enhance real-time healthcare operational intelligence systems

Conclusion:

This project demonstrates how healthcare analytics, forecasting, and visualization technologies can improve humanitarian operational awareness and support data-driven policy decisions within the UAC healthcare ecosystem.